

YENKATARAJALAXMIMANO HAR MEDA

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4 years of experience in software development and production-ready AI systems. Skilled in building backend services, GenAI applications, RAG pipelines, ML APIs, and cloud-native AI platforms. Passionate about learning new AI technologies and applying them to build scalable, secure, and real-world solutions using Python, FastAPI, Spring Boot, LLMs, Docker, Kubernetes, CI/CD, and MLOps.

EDUCATION

Saint Louis University

Master of Science, Computer Science — GPA: 3.98/4.0

Saint Louis, MO

Aug 2023 – May 2025

Lovely Professional University

Bachelor of Technology, Electrical and Electronics Engineering — Percentage: 89%

Punjab, India

Aug 2018 – Aug 2022

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, JavaScript, TypeScript

Machine Learning & AI: Supervised Learning, Classification, Regression, Feature Engineering, Model Training, Model Evaluation, Scikit-learn, Pandas, NumPy, Statistical Modeling, Time-Series Analysis

Deep Learning & GenAI: Neural Networks, CNNs, RNNs, LSTMs, Transformers, BERT, GPT Models, LLaMA, Prompt Engineering, Embeddings, Hugging Face, ChatGPT APIs, Google Gemini, Amazon Bedrock

LLM Applications & RAG: LangChain, LangGraph, RAG Pipelines, Vector Databases, FAISS, Pinecone, ChromaDB, Semantic Search, Intelligent Search, LLM Evaluation, Prompt Tuning, PEFT, LangSmith, LangFuse

Backend & API Development: FastAPI, Flask, Spring Boot, Spring Security, Spring Data JPA, REST APIs, Microservices, Hibernate, SQLAlchemy, Node.js, Express.js

Cloud & MLOps: AWS, AWS SageMaker, EC2, S3, RDS, Lambda, API Gateway, Docker, Kubernetes, MLflow, CI/CD, Jenkins, GitHub Actions, Terraform, Prometheus, Grafana

Data Engineering: Apache Kafka, RabbitMQ, PySpark, Airflow, ETL Pipelines, AWS Glue, Snowflake

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Snowflake, Amazon RDS, S3

Frontend Technologies: React.js, HTML, CSS, JavaScript, TypeScript, Bootstrap

Testing & Tools: Git, GitHub, GitLab, Bitbucket, Postman, Swagger/OpenAPI, JUnit, Mockito, Pytest, Unit Testing, Agile/Scrum

EXPERIENCE

Siri Consultancy Services

Remote

Software Engineer — AI/ML & GenAI Systems — Prior Authorization & Claims Intelligence GenAI Platform Feb 2026 – present

- Developed a GenAI-powered healthcare platform for claims review and prior authorization support.
- Built FastAPI and Spring Boot microservices for ML inference, LLM orchestration, document processing, and claims workflow APIs.
- Implemented RAG pipelines using embeddings, vector databases, chunking, metadata filtering, and prompt engineering.
- Integrated LLMs and ML models for clinical document summarization, claim classification, entity extraction, and missing document detection.
- Built agentic workflows to automate multi-step tasks: policy retrieval, document analysis, validation, and human reviewer handoff.
- Deployed AI services using Docker, Kubernetes, CI/CD, MLflow, and cloud infrastructure.
- Added production controls: PHI masking, RBAC, audit logs, monitoring, prompt safety, and responsible AI guardrails.

SYNTACTECH INC

Remote

Java FullStack Developer

Jul 2025 – Feb 2026

- Designed and developed multi-tier applications including presentation layer, business layer, and service layer components following enterprise Java best practices.
- Developed and exposed RESTful APIs using Spring Boot to integrate multiple systems with compliance requirements and deployed services to cloud environments.
- Built and consumed Java-based APIs with Spring Boot, handling service requests through Spring MVC Controllers and implementing backend logic using Java Collections.
- Designed and developed J2EE components and prepared technical design documents; reviewed and evaluated peer code to ensure quality and consistency.

- Created stored procedures and database interfaces as required for various work packages and projects.
- Developed web-friendly UI elements using Angular templates and integrated them with backend Spring Boot services.
- Built advanced applications leveraging JavaScript technologies and maintained high-quality production standards across all software deliverables.

Open Source Project (Raspberry Pi / Micronaut)

Saint Louis, MO

Lead Developer

Aug 2024 – May 2025

- Upgraded a Java-based Raspberry Pi hardware interaction project from Micronaut 3.7.3 to 4.7.6, resolving compatibility issues and enabling support for latest platform features.
- Developed comprehensive unit tests to validate GPIO functionality, increasing test coverage by 40%.
- Maintained the GitHub repository (issue triaging, code reviews, pull request management), improving development cycle efficiency by 20%.
- Mentored 5+ contributors on hardware setup, Java development, and Micronaut best practices, accelerating onboarding efficiency by 50%.

Saint Louis University

Saint Louis, MO

Teaching Assistant — Deep Learning

Aug 2024 – Dec 2024

- Assisted students with deep learning concepts including neural networks, CNNs, RNNs, backpropagation, optimization, regularization, and model evaluation.
- Supported hands-on labs and assignments using Python, PyTorch, TensorFlow, NumPy, pandas, and Jupyter Notebook for model development and experimentation.
- Guided students in debugging training issues, improving model performance, analyzing loss curves, and understanding practical AI/ML workflows.
- Reviewed coursework, provided technical mentoring, and supported the professor in delivering deep learning concepts through labs, projects, and student discussions.

ValueLabs Solutions

Hyderabad, India

Software Development Engineer (SDE)

Jan 2022 – Aug 2023

- Migrated REST APIs from Java JAX-RS to Spring Boot, enabling support for multiple PostgreSQL databases and improving system efficiency by 30% in a unified microservices architecture.
- Refactored and enhanced a Java & Selenium automation framework using keyword-driven and data-driven testing, improving test coverage and maintainability.
- Rebuilt a reporting dashboard by migrating a Ruby on Rails application to Node.js + Express.js, achieving better performance and responsiveness.
- Assisted in migrating on-premises infrastructure to AWS (EC2, S3, RDS) and implemented CI/CD pipelines with Jenkins, cutting deployment time by 50%.
- Optimized complex PostgreSQL queries and schema designs, reducing query execution time by up to 40% through efficient joins, partitioning, and query plan analysis.

PROJECTS

AI-Powered Hospital Support Bot | *GenAI, RAG, LLM, EHR Integration*

Dec 2025 – Feb 2026

- Developed a hospital support bot to capture tickets instantly from chat, email, and portal.
- Automated issue understanding, prioritization, and routing for faster incident resolution of common IT issues (login failures, printer issues, Wi-Fi drops, application crashes).
- Implemented runbook-based auto-resolution and escalated unresolved cases to the right support team with summaries.
- Supported ER-style triage workflows by collecting symptoms, vitals, allergies, and medication details.
- Integrated EHR-related context: patient details, lab reports, prescriptions, and treatment plans; reduced IT downtime and improved support efficiency.